

When a novice
diver swam with
some of the
ocean's most
feared predators,
he came away
with a new
appreciation
for them

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*A diver keeps a close watch on a
tiger shark in the Bahamas*

ADVENTURE



Swimming with **Sharks**

I saw *Jaws* the year it came out, 1975. I was nine, and I still remember how the theatre erupted when Brody finally killed the monster shark. I loved the movie, and that night I dreamed of a shark swimming up through the toilet and coming after me down the hall.

So when I got this assignment, I decided to do what I'd never wanted to do: swim with sharks. I would go to a place in the Bahamas known as Tiger Beach, where I'd dive with tiger sharks, the species responsible for more recorded attacks on humans than any shark except the great white. Most people who got wind of this plan thought I was either very brave or very stupid.

But I just wanted to puncture an illusion. The people who know sharks intimately tend to be the least afraid of them, and no one gets closer to sharks than divers. The divers who run operations at Tiger Beach give them nicknames and light up when they talk about their personality quirks. In their eyes these sharks aren't man-eaters. (In 2017 there were just five fatalities from shark attacks worldwide.)

But tiger sharks are not relevant just because of how many people they bite. As apex predators, they act as a crucial balancing force in ocean ecosystems, constraining the behaviour of other animals. As such, they are essential to the health of sea

grass ecosystems, habitat to marine wildlife.

Furthermore, their role in ocean ecosystems is likely to increase if the planet and its oceans continue to warm. Tiger sharks love warm water, they eat almost anything, and they have large litters of pups, making tigers one of the hardiest shark species. They are also among the largest: mature females can exceed 5.5 metres and weigh more than 540 kilograms. Only great whites and a few other shark species are larger.

TIGER BEACH IS NOT actually a beach. It's a shallow bank about 40 kilometres from Grand Bahama Island, a patchwork of sand, sea grass, and coral reef that began attracting divers about a decade ago. It's prime habitat for tiger sharks and has ideal conditions for viewing them. The water is six to 13.5 metres deep and usually crystal clear. You strap on weights, sink to the bottom and watch the sharks go by.

On the boat ride to the site, our dive operators, Vincent and Debra Canabal, started tossing bloody chunks of fish overboard. Almost immediately the water filled with Caribbean reef sharks and lemon sharks. At last Vincent spotted a huge dark silhouette. "Tiger!" he yelled, pointing.

He jumped in with a crate of mackerel to begin feeding the shark on the seafloor – in part to occupy it while the rest of us entered the water, and



With jaws and teeth designed to crush and shear hard objects such as turtle shells, a tiger shark can afford to bite first and worry about edibility later

in part to make sure it wasn't too hungry when we did. All of this was OK with me until I reached the bottom and immediately had to fend off the first tiger shark I'd ever laid eyes on, all 360 kilograms of it.

The way Debra described it later, this was just 'Sophie' being curious and friendly. "She *loooved* you," Debra said, because of all the attention Sophie paid me during the dive. At the time I wasn't sure.

But after watching how Vincent and Debra handled them over the next week's dives – caressing them after feeding them a fish, steering them gently away when it was time for them to move on – it became easy to see the sharks in a very benign light. Not once did they make a sudden or aggressive move towards anyone; they moved slowly and

deliberately, swimming in large loops and then coming on a glide path to the feeding box. I felt surprisingly safe in their presence.

Most of the tiger sharks at Tiger Beach are habituated to divers, used to being fed. But even the ones that aren't familiar with the routine generally are not dangerous to divers. Tiger sharks are ambush predators, relying on stealth and surprise to catch their prey.

At Tiger Beach you're not blindly paddling or swimming at the surface, like most attack victims. You're down at the sharks' level, presenting yourself as something other than prey, and reasonably safe.

However, there are videos of near misses at Tiger Beach – in one a tiger goes after a diver's leg and in 2014 a diver simply disappeared.

Our group even had a scare when an angelfish wandered into our midst and the reef sharks and lemon sharks went into a frenzy, chasing it as it hid between people's legs. Everyone thought someone was going to be bitten in the melee, and there were three 450-kilogram tiger sharks milling around that might suddenly have taken an interest in a wounded diver.

We were back in the water the next day, but it was the kind of fluke that reminds you that sharks are wild and inherently unpredictable. And according to scientists who study them, tigers are especially unpredictable.

I FLEW TO OAHU to meet Carl Meyer at the University of Hawaii to discuss his research on the recent spike in tiger shark attacks in Hawaii - which have jumped from an average of just over four a year from 2000 to 2011 to more than ten a year in 2012, 2013 and 2016. Meyer and his team have tagged hundreds of tiger sharks with satellite tags and acoustic tracking devices. He says they're just beginning to understand them.

"Tiger sharks can show up any time of day or night, and they may be there one day and back the next day, or gone for three years," he says.

At least some of this unpredictability is likely caused by the sharks'



A newborn pup has the striped markings along its metre-long body that give the tiger shark its name. The markings fade as the shark grows

hunting habits, he says. Tiger sharks rely on surprise to catch their prey. "If you're predictable, your prey is going to adapt to that predictability. It makes sense to suddenly appear in an area and not be there very long."

The uptick in attacks in the autumn might also be due to having more sharks around the islands at that time of year. That's when tiger sharks come in to give birth, Meyer points out. Female tiger sharks make a huge energy investment when they ovulate. Their eggs are the size of baseballs, and they can have 80 pups in a litter. That might mean that female sharks reach the island hungry and needing to replace energy reserves after giving birth.

Another possible theory involves a proliferation of sea turtles. After decades of intense exploitation, green sea turtles received protection in 1978. They're now common off

Hawaii's shores and are a familiar food for tiger sharks. With wide jaws and heavy, angled teeth, tiger sharks are able to crush and slice through an adult turtle's shell in a way most sharks can't. So if more turtles are sharing the water with more people, more shark bites might be the result.

The relationship between tiger sharks and sea turtles could have broad implications for the health of ocean ecosystems around the globe.

On a remote part of Australia's western coast called Shark Bay, a research team led by Mike Heithaus of Florida International University has documented how tiger sharks prevent sea turtles and dugongs (sea cows) from overgrazing the sea grass beds that anchor the ecosystem. It's not just by eating the animals, they discovered. The presence of the sharks changes the turtles' and dugongs' habits, forcing them to graze more judiciously to lessen their risk.

In the Bahamas, which designated its waters a shark sanctuary in 2011, the marine ecosystems are relatively healthy. But the adjacent western Atlantic has much weaker shark protections and appears to be suffering the consequences.

"I do work in Florida and the Bahamas, and it's night and day," says Neil Hammerschlag, a marine ecologist at the University of Miami. "We see massive differences in the size and

numbers of the sharks. They're doing well in the Bahamas, but we almost never catch them off Florida. They're just 80 kilometres apart," he added. Florida prohibited the killing of tiger sharks in its waters in 2012, but it's the only state on the United States' eastern seaboard to have done so.

JAWS ISN'T RESPONSIBLE for most of the threats tiger sharks face – coastal development, marine pollution, longline fishing, the popularity of shark fin soup – but it did create a cultural attitude that has had a long shelf life. After *Jaws*, people didn't just become paranoid about sharks; they became callous, even vengeful. Shark-fishing tournaments sprouted on the eastern seaboard of the United States, and dozens continue, celebrating the spectacle of 'monster sharks' hanging on the docks.

In summer 2015, as I was planning my dive at Tiger Beach, news broke that an 360-kilogram tiger shark had been caught off the South Carolina coast. *USA Today* called the shark "monstrous" and described the fishermen as "brave". When I got home from Hawaii, I looked at the story again. Seeing the picture of the gutted, deflated shark, I thought about how it was once the same size as Sophie, and those weren't at all the words that came to mind – for either the shark or the men who killed it. **R**